

Andrea Valente

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Summary

AI Computer Vision Engineer working across model development, geometric vision and inference systems. Builds evaluation pipelines and practical tools that turn research findings into client decisions and operational R&D workflows.

Experience

Deep Vision Consulting

Modena, Italy

AI Computer Vision Engineer

Apr 2024 – Present

- **Package-seal inspection:** Became primary implementation owner of a Python analysis suite for a global packaging manufacturer, combining image segmentation, geometric measurements and force features.
- Designed thin-plate-spline defect synthesis and a tailored U-Net workflow, using source-aware splits and reflection augmentation to address scarce examples and misleading edge reflections.
- Built the cloud-integrated dashboard used by the sealing team on new samples as an operational R&D tool, with queued GPU tasks and failure inspection; supported client engineers with deployment and handover.
- **Efficient pose estimation:** Implemented Python training, inference, camera correction, evaluation and tracking for 6DoF head pose; comparisons supported joint selection of a 0.6 MB LCNet model versus a 2.4 MB reference, with equivalence within client-agreed rotation/translation thresholds.
- Integrated ONNX Runtime with WebGPU, graph capture and GPU preprocessing into a pre-production test application; profiled runtime behaviour and tested across high-, mid- and low-tier devices.
- **3D reconstruction:** Created the client's first quantitative reconstruction framework for a global luxury eyewear group, annotating and fitting 68 face scans to FLAME and defining region-specific mesh errors and eyewear-relevant measures.
- Designed the acquisition protocol and prepared a 48-subject, 7,608-take benchmark; findings on 3D within-take stability and 2D across-session reproducibility informed a joint client/team decision to prioritise more viable facial-measurement approaches.
- **Point-cloud geometry:** Developed Python RANSAC-based fitting and synthetic-data tools for smart-lighting point clouds; the package was evaluated on selected shapes through visual feedback.

AI Computer Vision Engineer (Intern)

Sep 2023 – Mar 2024

- Evaluated NeRF/MVS and deformable meshes against virtual try-on acquisition and computation constraints; generated approximately 60,000 synthetic face images and trained a shape-regression network, identifying scale ambiguity and synthetic-to-real limitations.

Vaia Innovations

Remote, part-time

Frontend Web Developer

Jun 2022 – Sep 2022

- Built responsive interfaces from Figma designs for article exploration and writing on a Web3 publishing platform, using TypeScript, Vue.js and Tailwind CSS.

Skills

ML and evaluation: Python, PyTorch, OpenCV, NumPy, MLflow; synthetic data, segmentation, model training and source-aware validation.

Geometric vision: 3D reconstruction, FLAME mesh fitting, camera geometry, point-cloud processing, RANSAC, SciPy optimisation, Open3D, PyTorch3D, Blender.

Inference and testing: ONNX Runtime, WebGPU, TFLite, GPU profiling, Perfetto, Playwright, cross-device testing.

Software and delivery: Git, Docker, Celery, RabbitMQ, PostgreSQL, cloud-storage integration; TypeScript, Vue.js, Tailwind CSS.

Education

MSc in Artificial Intelligence — University of Bologna

110/110 with honours

Sep 2021 – Mar 2024

BSc in Computer Science — University of Bologna

110/110 with honours

Sep 2018 – Dec 2021

Languages

Italian: native. English: professional working fluency; MSc taught in English.

IELTS Academic: overall band 7.0 (Jul 2021).